



सत्यमेव जयते

Ministry of Environment,
Forest and Climate Change
(MoEFCC)



SPARC FOUNDATION
INDIA

RESEARCH FORUM LECTURE TOPIC

Human - Wildlife Coexistence - Novel approach to connect by technology

Mr. Pradeep Sathapathi

CEO SPARC FOUNDATION

Ex- Project Researcher DRDO & Mission Director

3rd Year B.Tech | Dept. of ECE | College of Engineering

Hearty Welcome



Dr.B.R. Ambedkar University - Srikakulam,Etcherla

RESEARCH FORUM LECTURE SERIES



SPARC FOUNDATION

Andhra Pradesh, India

introducing

WILDTRAK 360

AI-Powered Prediction & Rapid Response Network for Human–Wildlife Coexistence



SPARC FOUNDATION
INDIA

PROJECT WILDTRACK 360

WILDTRACK

Mr. Pradeep Sathapathi

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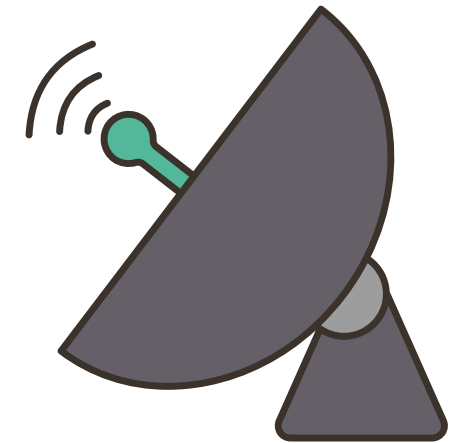
Ex- Project Researcher DRDO & Mission Director

ISRO| Dept. of Space - Space tutor partners



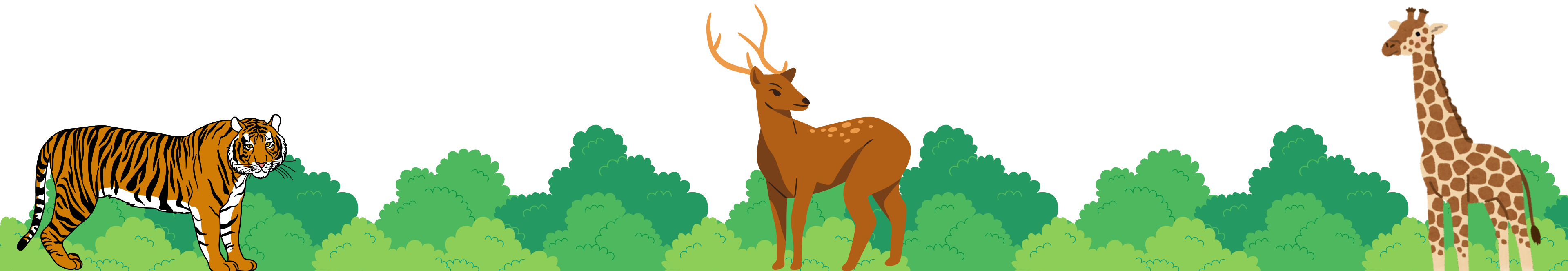
Introduction

WILDTRAK 360

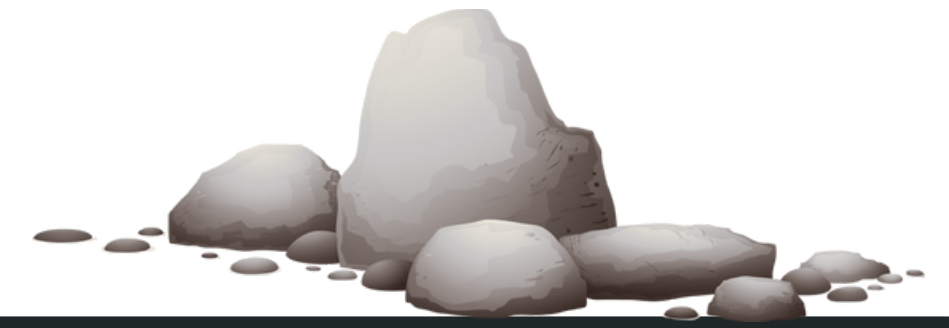


AI-Powered Prediction & Rapid Response Network for Human–Wildlife Coexistence

- Project Theme : Technology Solution



Problem Statement:



Identified Problems & Key Challenges – WildTrak 360

- **Escalating Human–Wildlife Conflict**

Crop destruction, property damage, and injuries are increasing in forest-edge villages, threatening both livelihoods and biodiversity.

- **Lack of Early Warning**

Communities rarely receive timely alerts when elephants, leopards, or other animals move toward farms or settlements.

- **Slow & Manual Reporting**

Current methods depend on phone calls or physical patrols, which often reach forest staff only after an incident has occurred.

- **Limited Real-Time Intelligence**

Forest departments lack continuous, on-ground data to predict animal movement or coordinate rapid response.

- **Technology Gaps in Rural Areas**

Many conflict zones have poor network coverage and limited access to advanced monitoring tools.

- **Risk to Both Humans and Wildlife**

Delayed action leads to injuries, retaliatory attacks on animals, and long-term ecological imbalance.

Project Description:



Human–wildlife conflict is growing across India, leading to crop loss, property damage, and danger to both people and animals. Villagers often get no advance warning, and forest staff usually arrive only after the damage is done. WildTrak 360 solves this by creating a three-layer protection system: solar-powered sensors on forest edges detect animal movement, an AI engine predicts risks using live data and past patterns, and drones with night-vision cameras track animals in real time. When a threat is detected, instant multilingual alerts are sent via WhatsApp, SMS, and a mobile app, giving communities up to an hour to prepare. This proactive system helps people stay safe, protects livestock, and allows forest officials and trained volunteers to guide safe responses without harming wildlife. Affordable, scalable, and community-driven, WildTrak 360 not only safeguards villages and animals but also creates local jobs and supports conservation.

Methods:

- **Site Assessment & Community Mapping:**

We begin by identifying high-conflict corridors in collaboration with forest officials and village leaders. Key areas such as farms, water sources, and communication lines are mapped to determine optimal sensor and drone deployment points. Awareness sessions are held, and local volunteers—called “Tech Rangers”—are recruited from students and farmers to participate in monitoring efforts.



- **Sensor Grid Deployment:**

Solar-powered motion and thermal sensors are installed along forest edges and known animal paths. Using low-power LoRaWAN or Wi-Fi, these units are calibrated to local wildlife patterns—like elephant height or leopard stride—to minimize false alarms.



- **AI-Powered Prediction & Monitoring:**

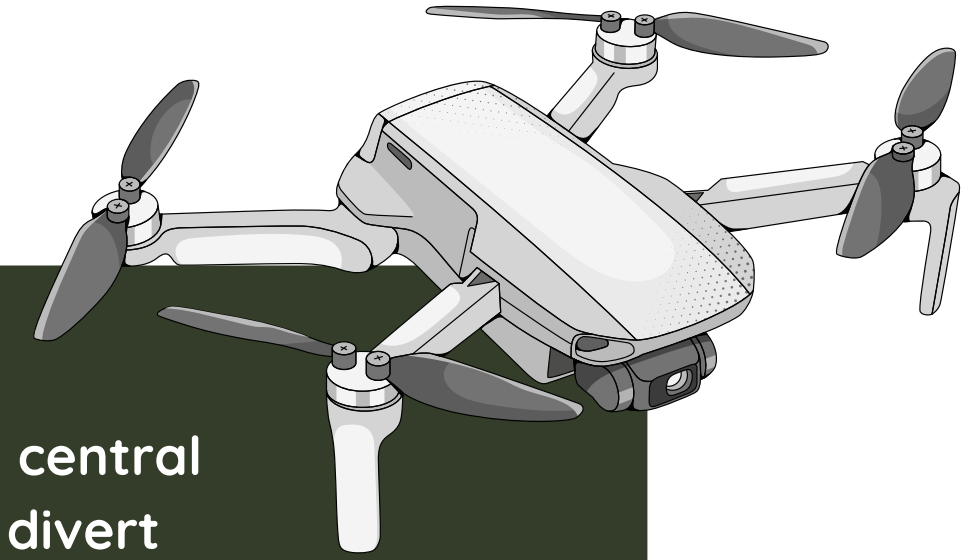
All sensor data, satellite imagery, and historical sightings feed into a lightweight AI engine built with models such as YOLOv8 or TensorFlow Lite. This system identifies species in real time and predicts movement paths 30–60 minutes ahead. Continuous field feedback from Tech Rangers and forest staff ensures the AI improves over time.



Methods:

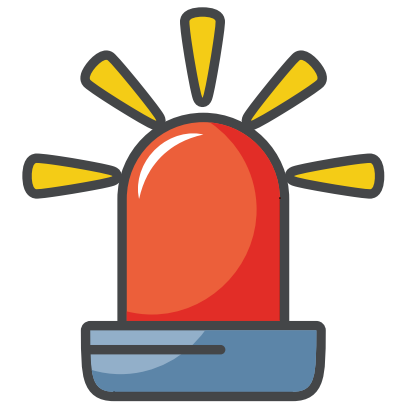
- **Drone-Assisted Rapid Response:**

Compact drones with infrared cameras are deployed automatically to hotspots, streaming live video to a central dashboard powered by tools like Streamlit and Google Maps API. Forest guards use these feeds to safely divert animals, tranquilize if needed, and track movement efficiently.



- **Real-Time Alerts & Community Engagement:**

When risks are detected, instant alerts are sent via WhatsApp, SMS, or a simple mobile app. Multilingual notifications include species, GPS location, and predicted route. In areas with poor connectivity, Bluetooth mesh networks or local sirens act as backup, helping villagers safeguard people and livestock.



- **Feedback & Continuous Improvement:**

After each event, ground reports and sensor logs are analyzed to retrain AI models, refine sensor placement, and improve alert accuracy. Insights are shared with state forest departments, enabling replication of the system in other conflict-prone zones.



Project Solution:

List of features offered by the solution

- The system incorporates radar-based tracking of unusual animal movement as an early indicator of potential forest fire events.
- Upon fire detection, automated alerts are sent instantly to the nearest forest stations, police stations and fire stations for rapid response.



CONNECT

with a Click

- featuring advancement in Safety

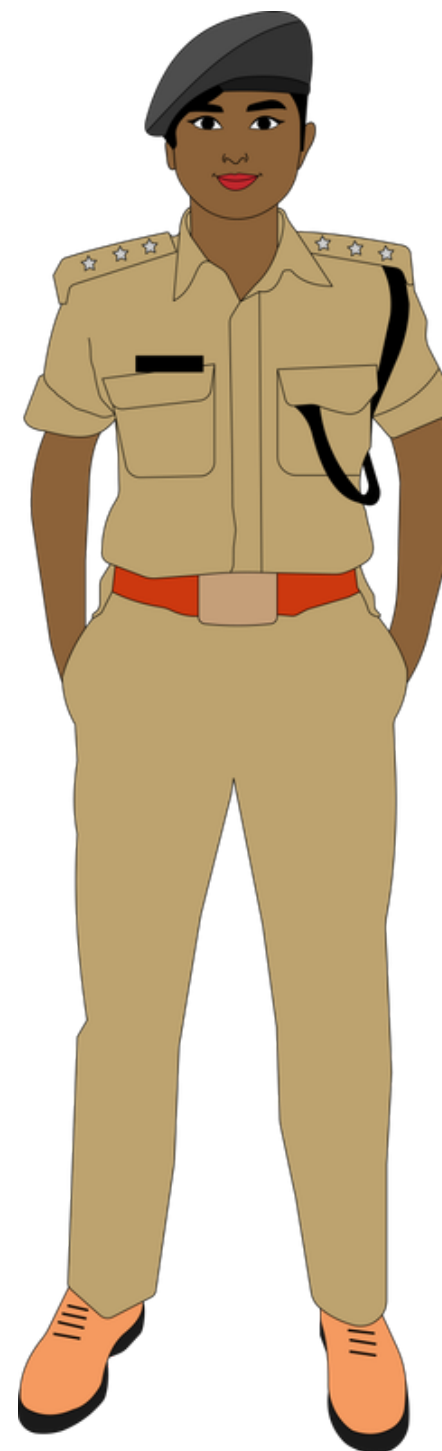




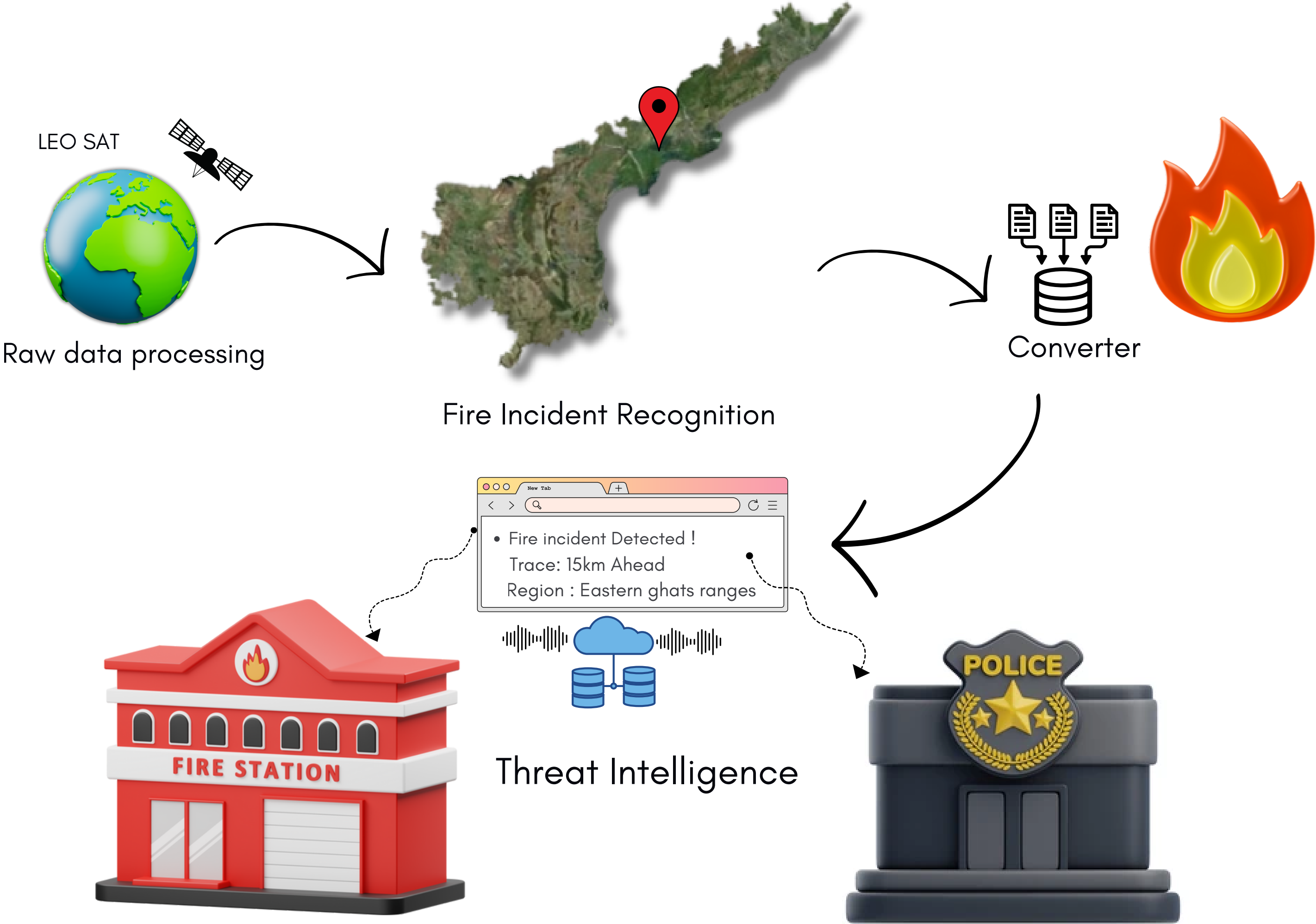
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(MoEFCC)

Introducing MISSION TRINETRA



Feature 1:



Feature 2: Threat Intelligence



Suspicious Moment



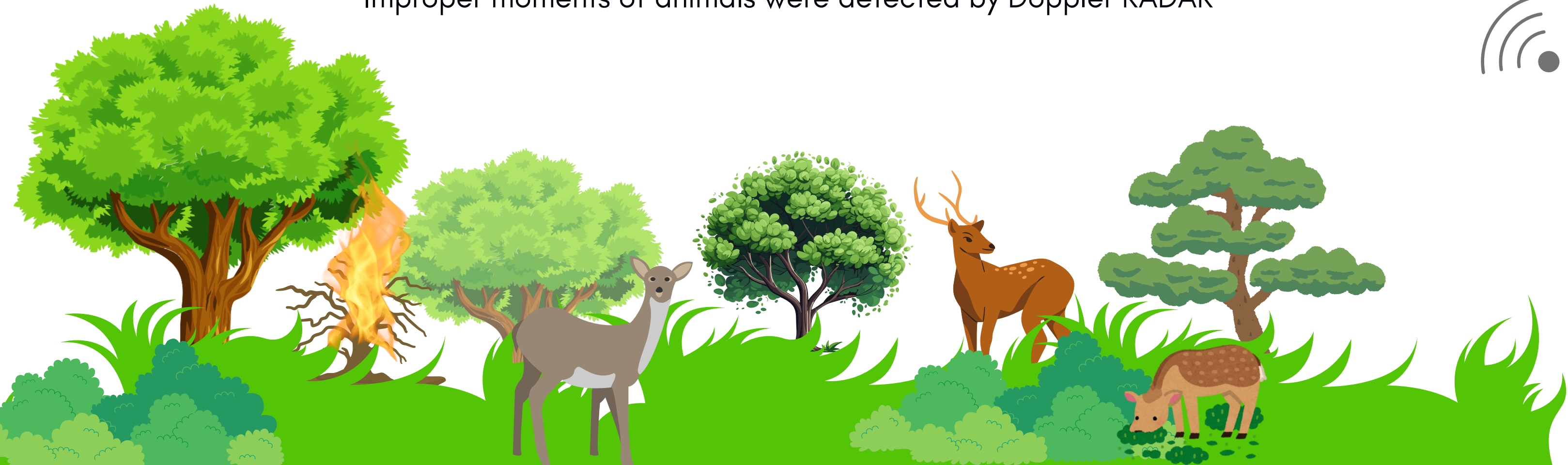
Improper Condition

Improper moments of animals were detected by Doppler RADAR

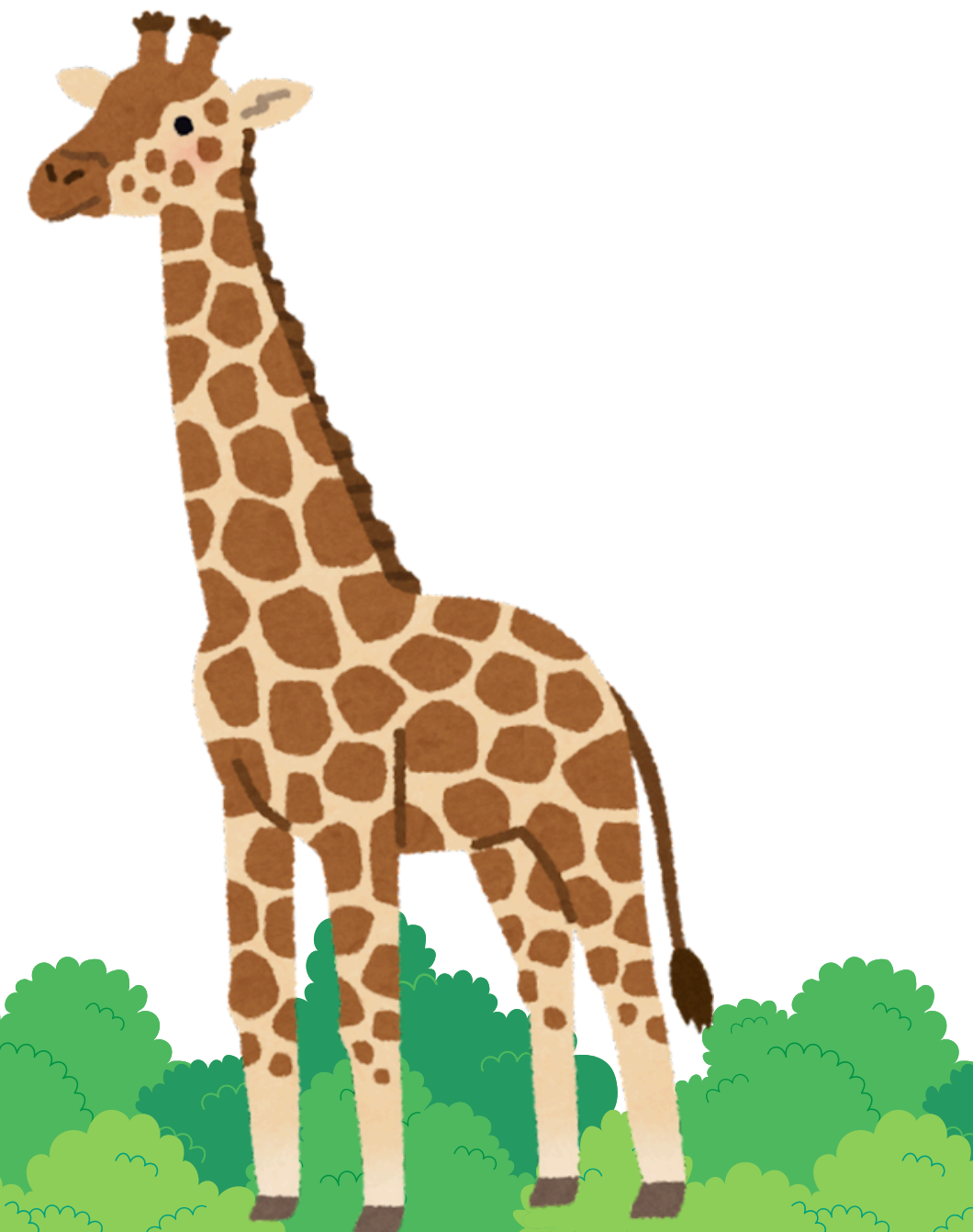
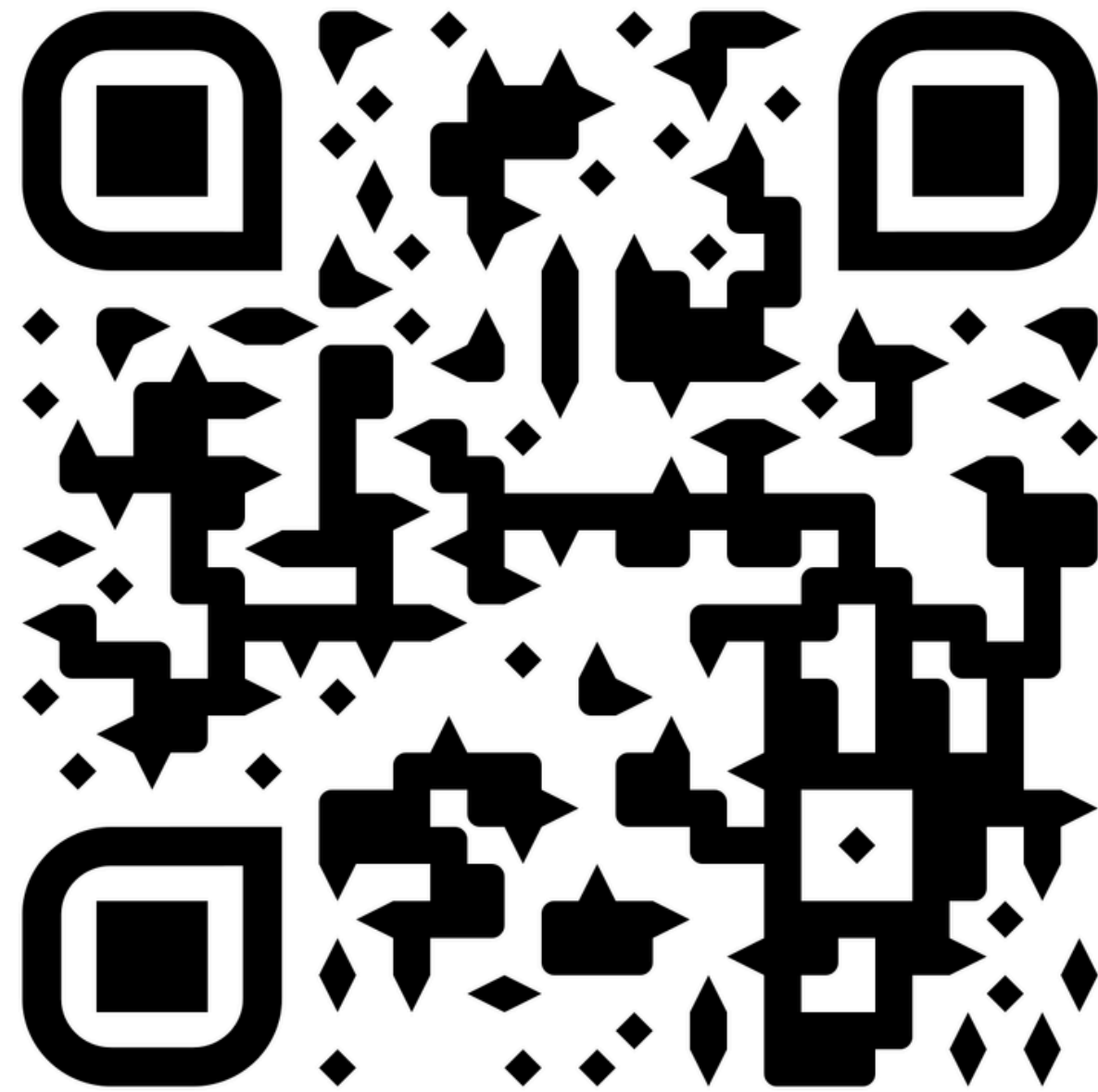
○ **ALERT**

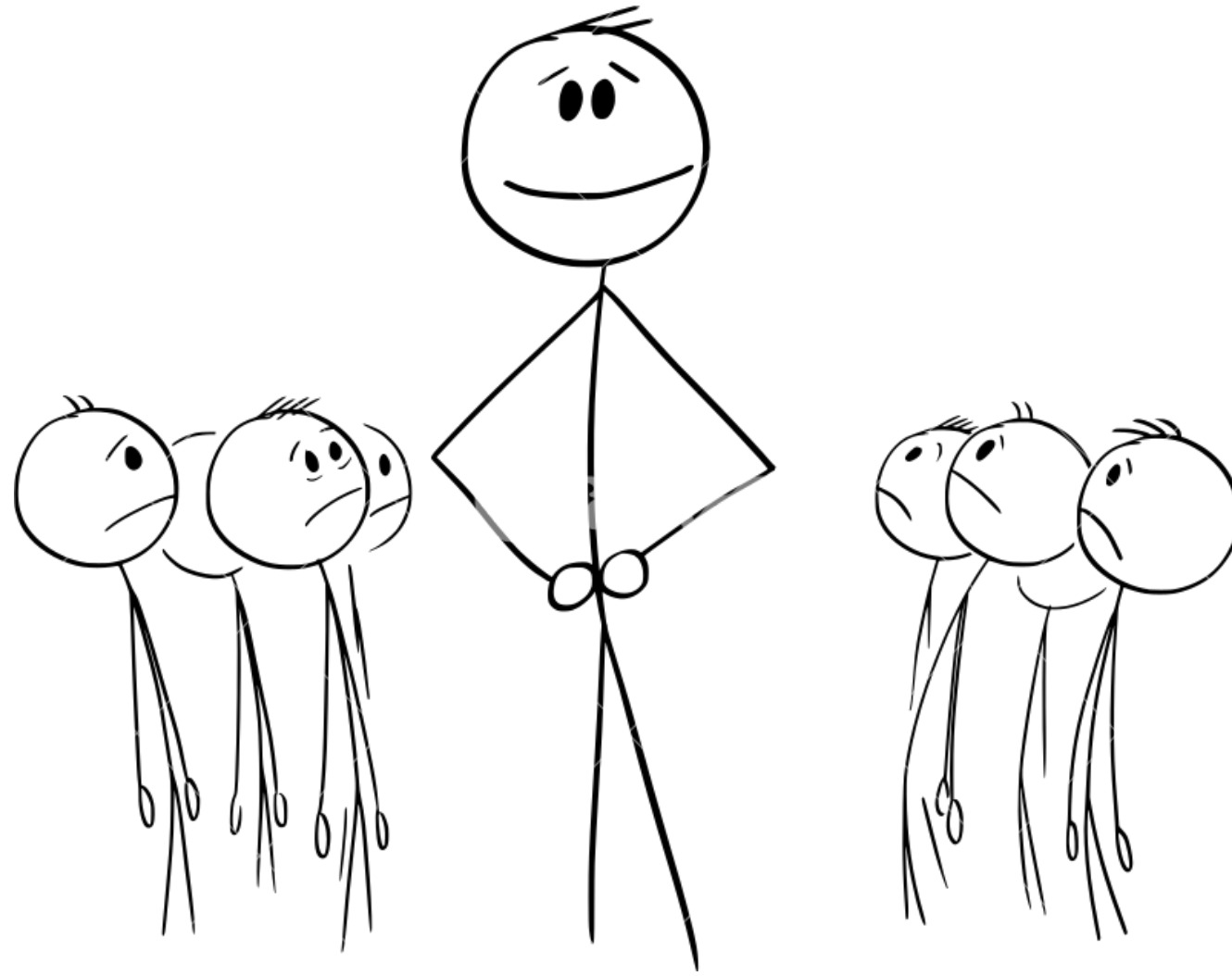
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- Suspicious Moment Detected !
- RADAR range: 30km
- Activity: Improper behavior



WILDTRAK 360

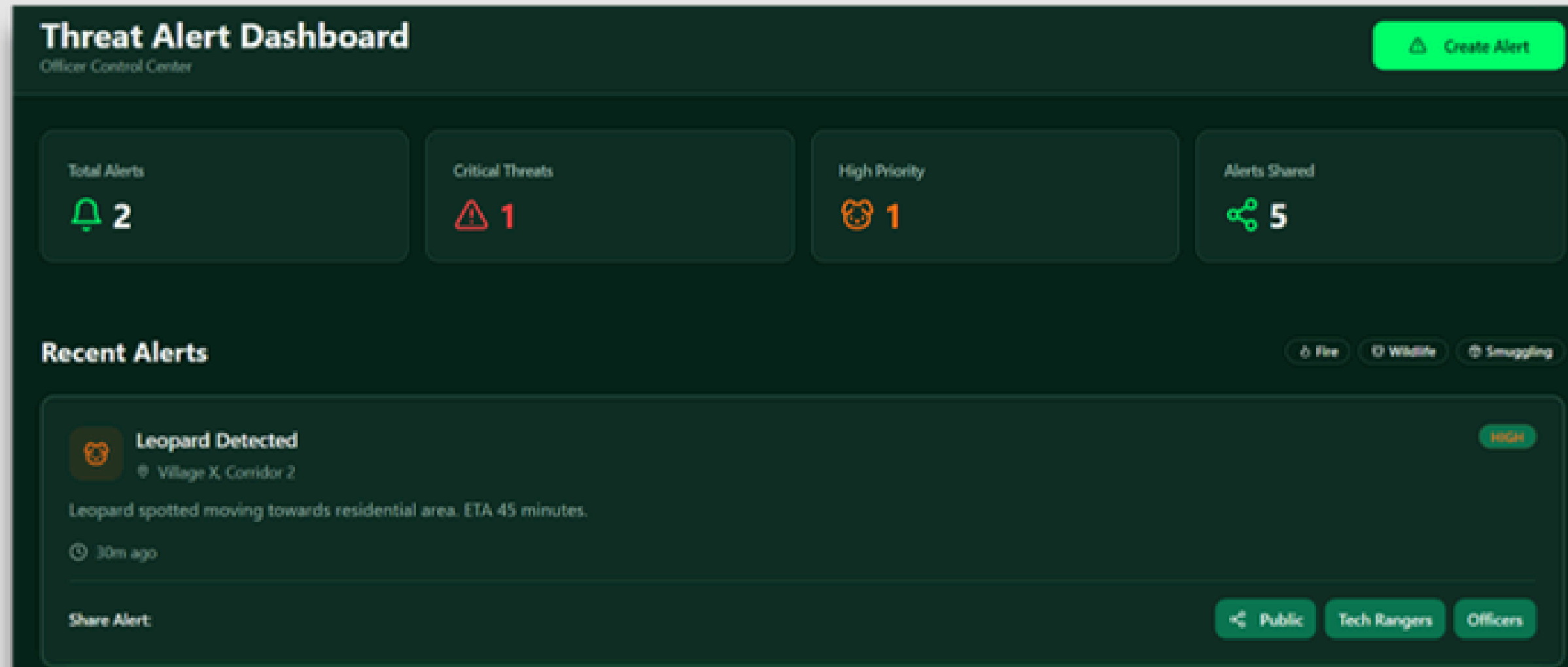




**What makes WildTrak 360
unique?**

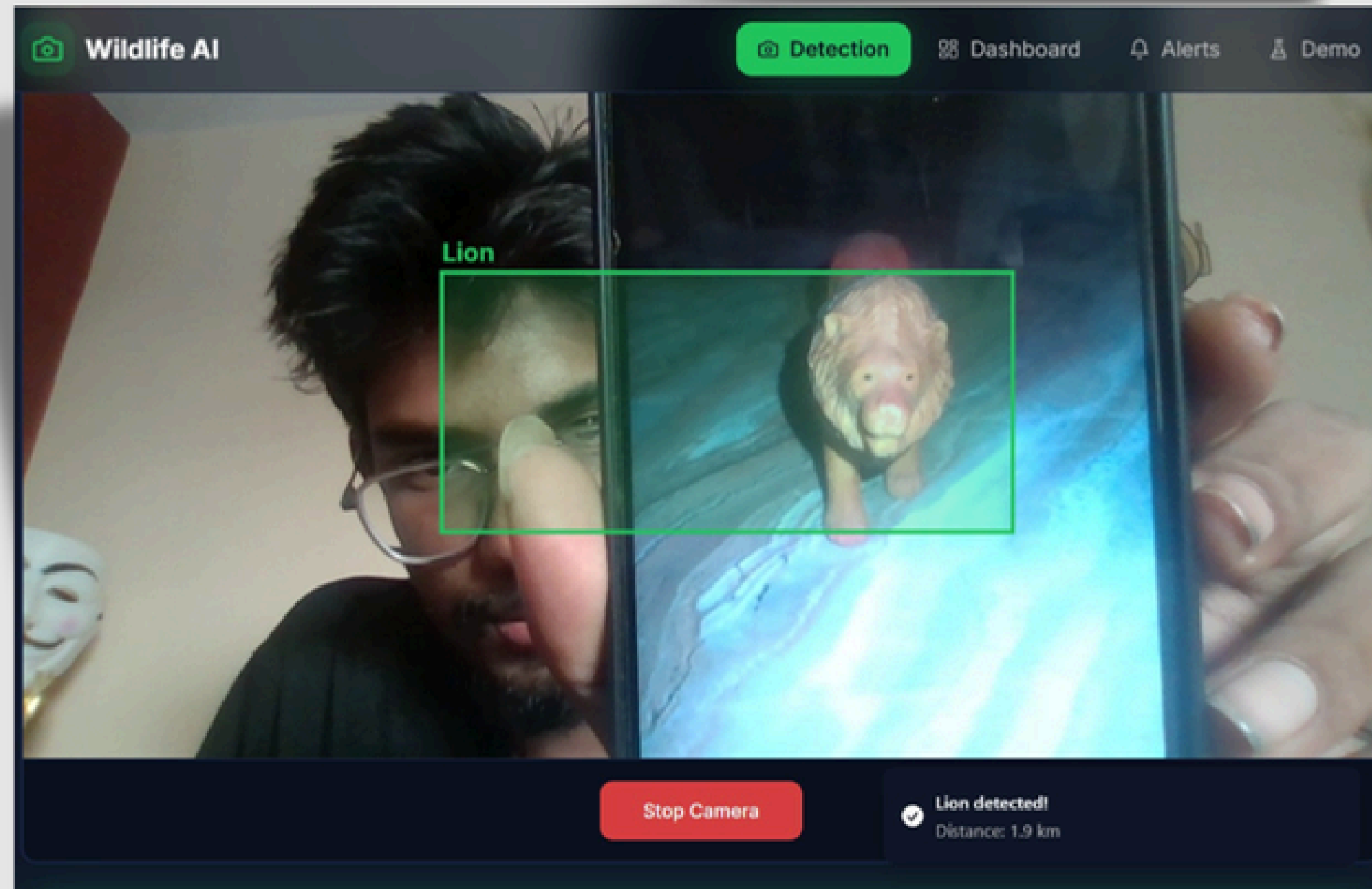
Community Knowledge Integration (Crowdsourced Wild Intelligence)





Empowers tribal communities as co-researchers,
strengthening human+AI synergy.

Innovation into **WILDTRAK 360**



What if there is no Internet?



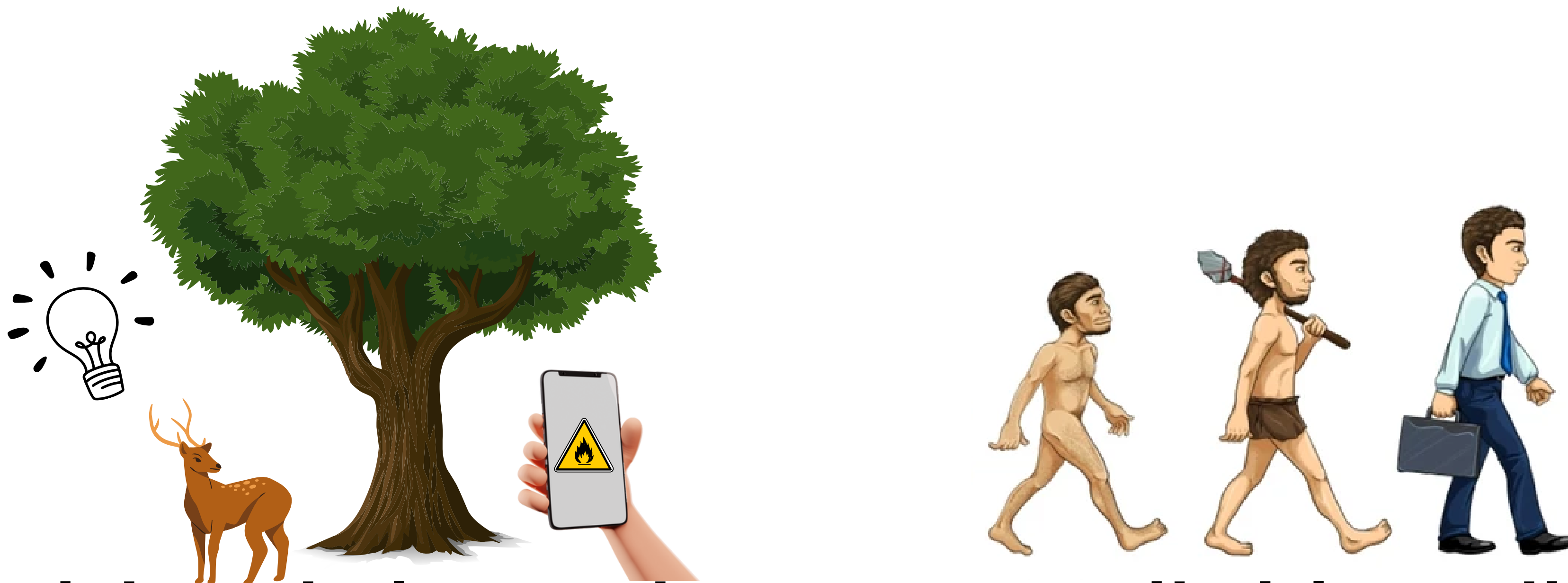
WILDTRAK 360



WILD SUPPORT

FM/AM Radio (Public Broadcast)

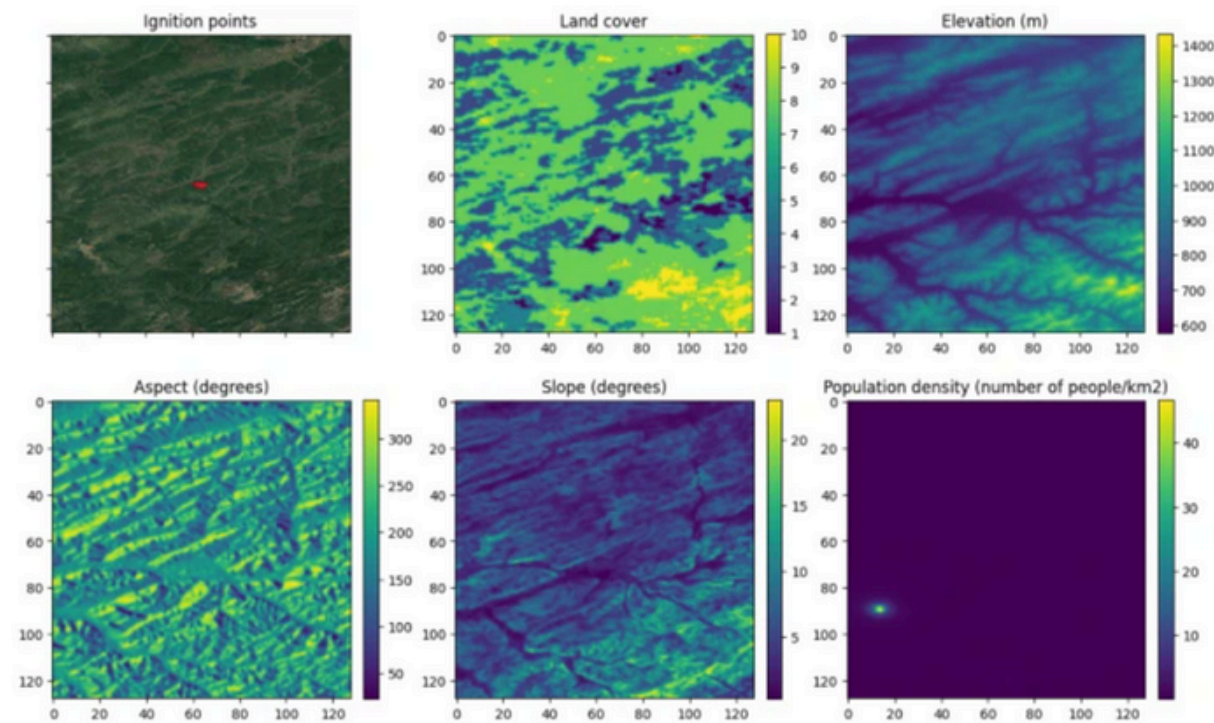
Works even in remote areas if there's a transmitter tower within 60–100 km



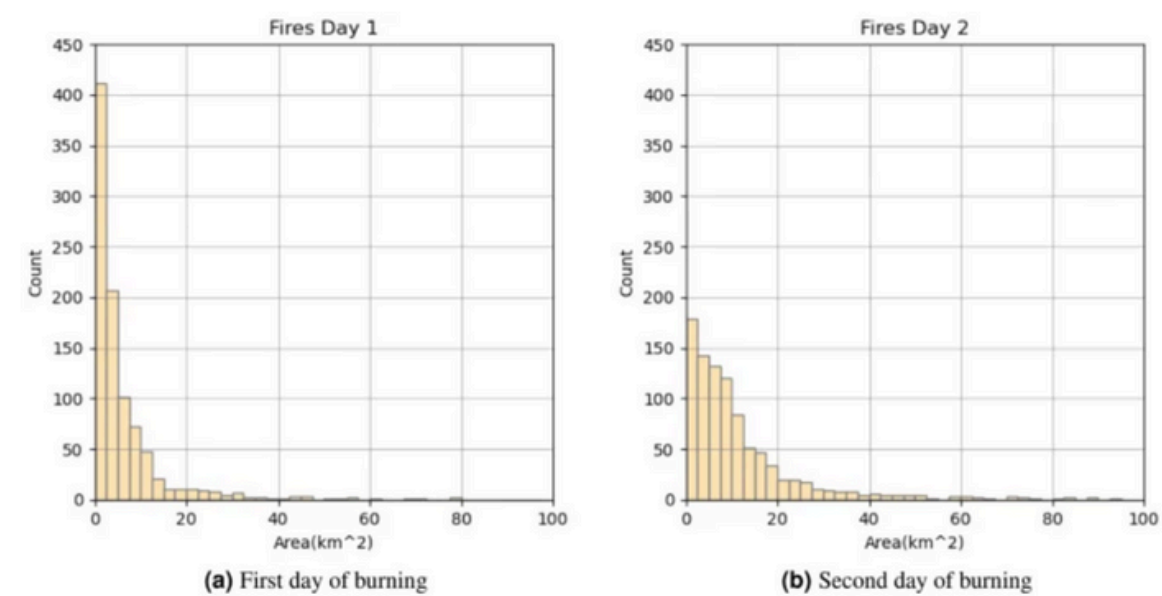
integrated ecosystem + community job creation

WILDTRAK 360

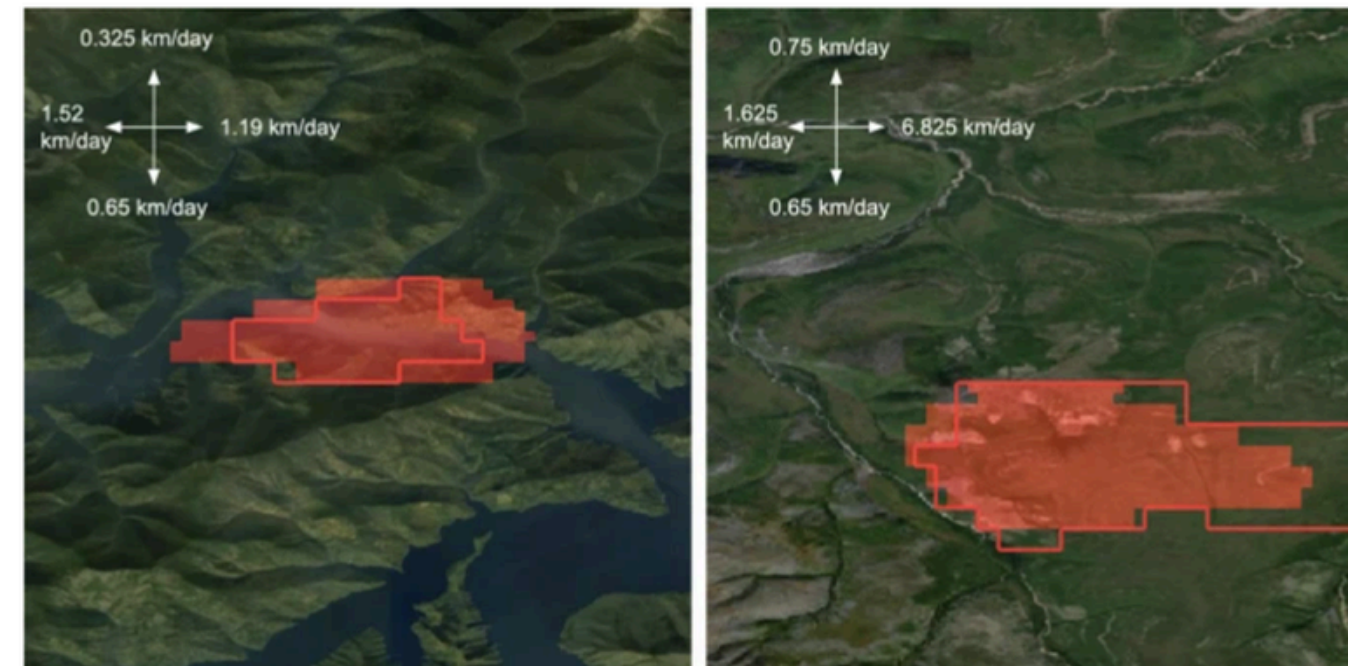
Glimpse of our prototype solution:



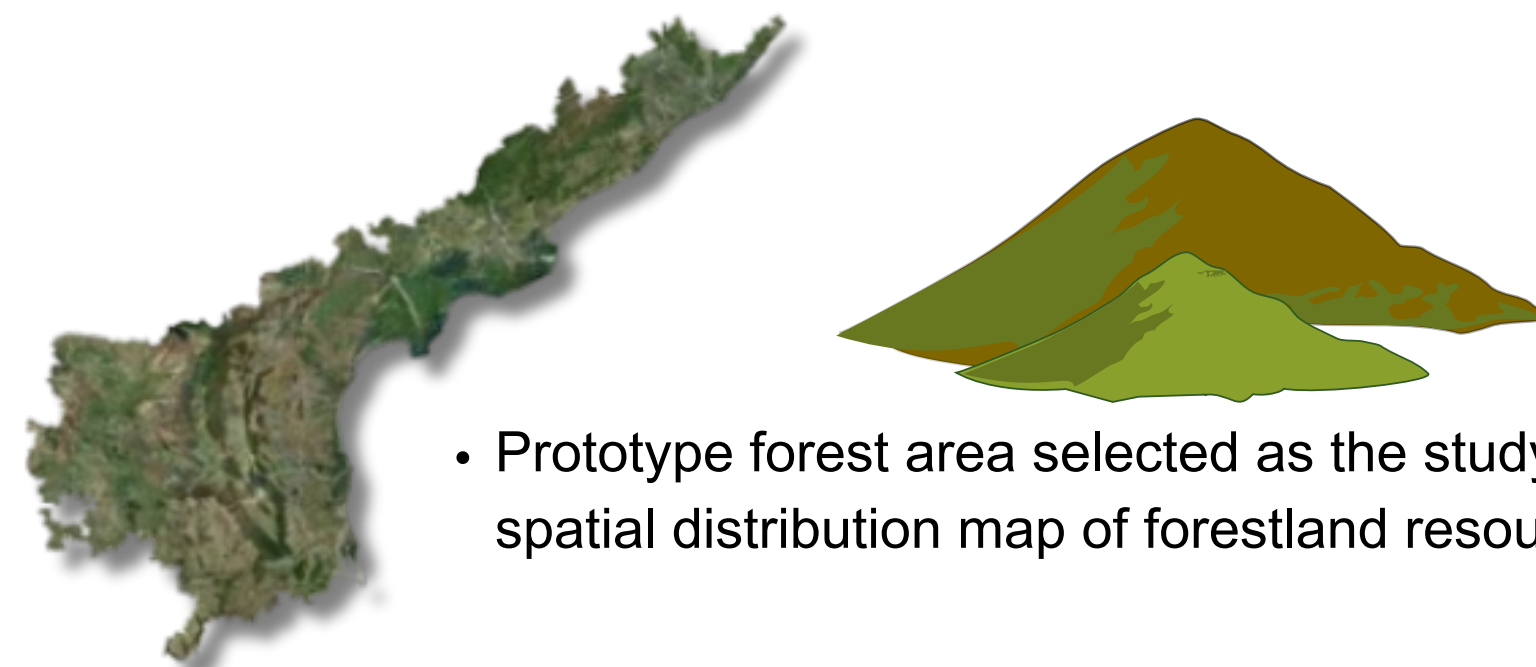
- Ignition points and images in “static” field



- Distribution of burned areas for the first 2 days of burning.



- Example of model predictions for three days on the test set (MA-Net model). The red boundary is for ground truth fire perimeter, the red area is for model's prediction.



- Prototype forest area selected as the study area and the spatial distribution map of forestland resources.

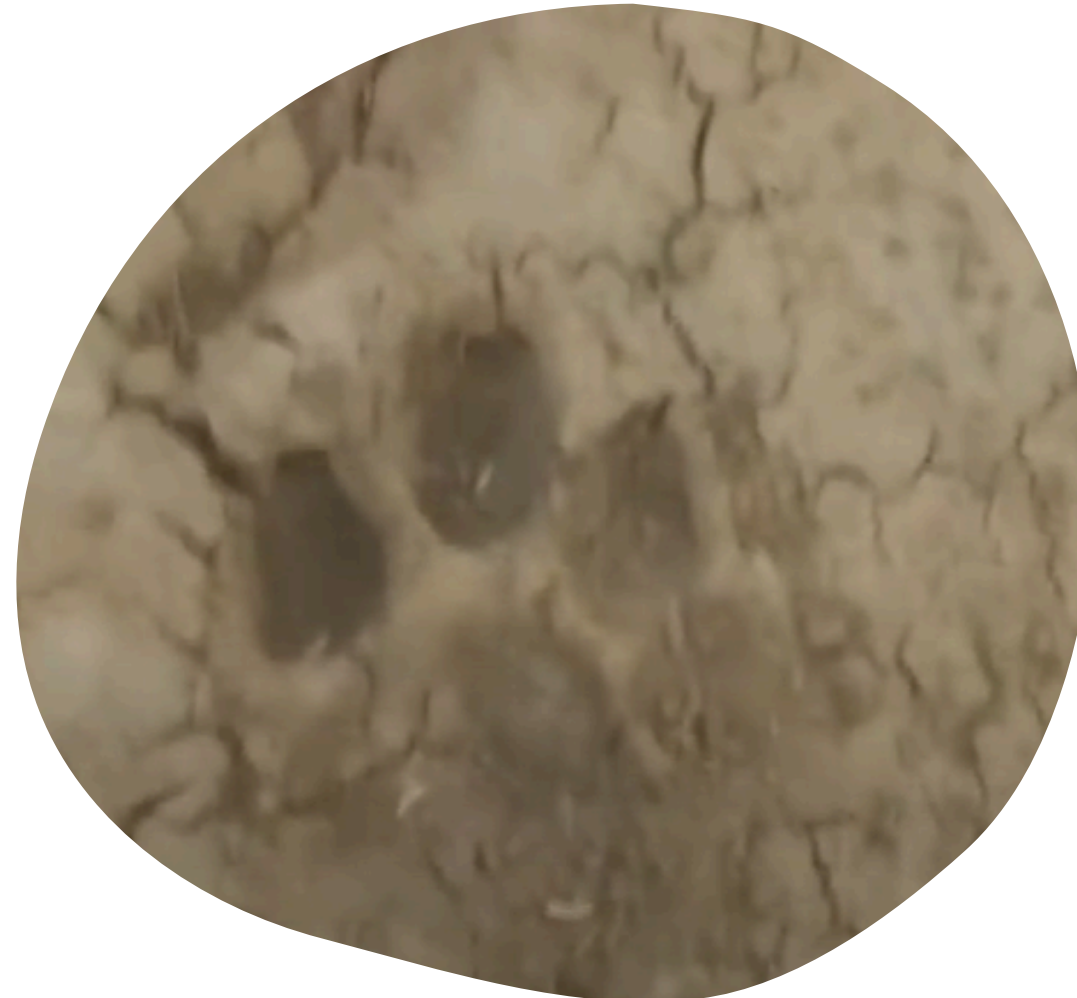
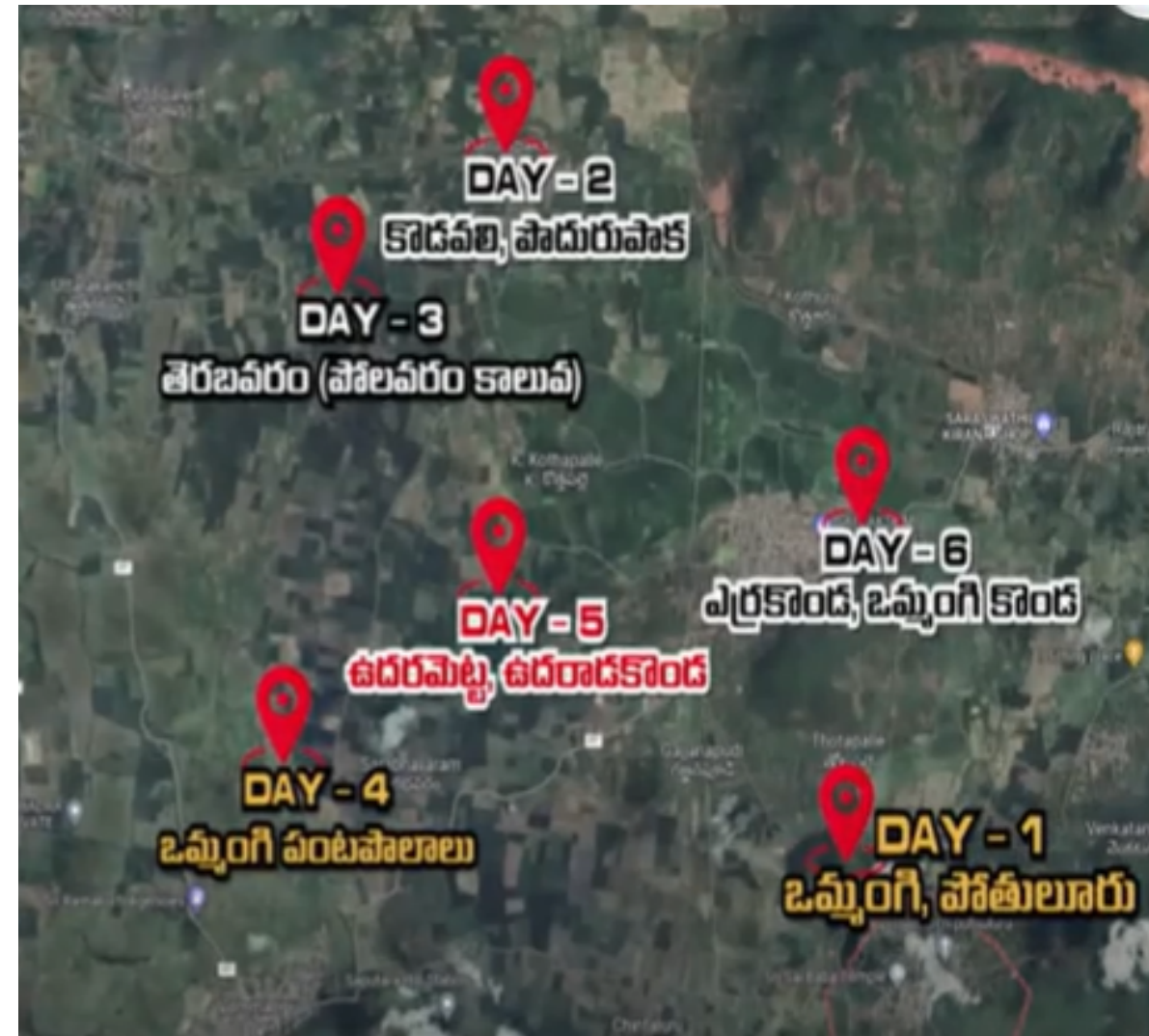
Case Study:

Yeleswaram Reserve Forest: Efforts to trap tiger continue

Pug marks of big cat found near Lododdipalem-Gowravaka villages.

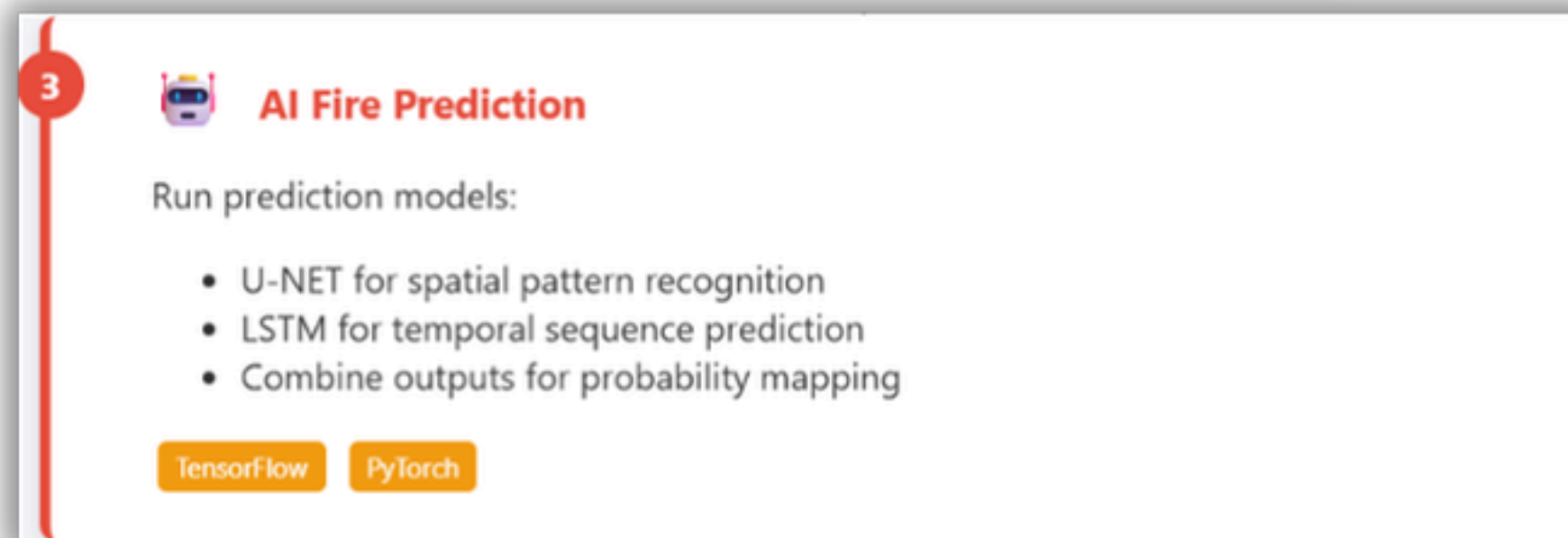
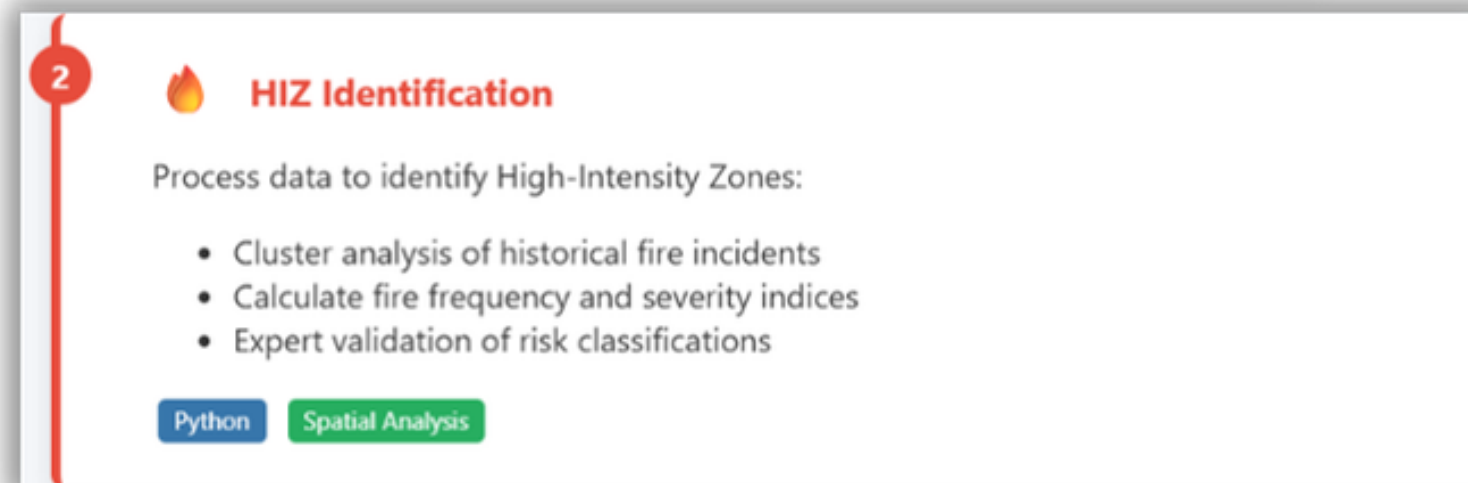


- KAKINADA: The male adult tiger reportedly entered the Lododdipalem-Gowravaka reserve forest, which is 6 km away from Serabhavaram. The pug marks of the tiger was found near Lododdipalem-Gowravaka villages. It is near the Patti Chinthala (Dharakaluva) project.
- Yeleswaram Reserve Forest Officer (RFO) J Srinivas said, citing reports, that the tiger is active from 5 pm to 6.30 am. The tiger was staying in Prathipadu mandal' reserve forest area. From May, it was spotted near human habitations. The big cat has been roaming around the reserve forest area near Lingamparthi, Vajrakutam, Pothuluru and Serabhavaram villages.
- Around 150 forest officials have been deployed to track and catch the tiger. As many as 52 CCTV cameras and five cages and three mobile cages have been set to trap the wild animal



Example Case:

Process flow diagram or Use-case diagram



WILD THREAT FIRE PREDICTION & SIMULATION SYSTEM

flow diagram



4



Radar Monitoring

Supplemental early detection:

- Monitor wildlife movement patterns
- Detect anomalous behavior signaling fire
- Update prediction models in real-time

Radar Tech

Anomaly Detection

5



Fire Spread Simulation

Dynamic modeling of fire progression:

- Cellular Automata (CA) modeling
- Simulate at 1, 2, 3, 6, 12 hour intervals
- Include real-time wind and terrain factors

CA Modeling

Raster Analysis

6



Output & Visualization

Generate actionable outputs:

- Color-coded probability maps (30m resolution)
- Animated fire spread simulations
- Risk assessment reports

WebGIS

Matplotlib

7



Early Warning System

Alert and response mechanisms:

- Automated alerts for high-risk areas
- Suggested containment strategies
- Evacuation route planning

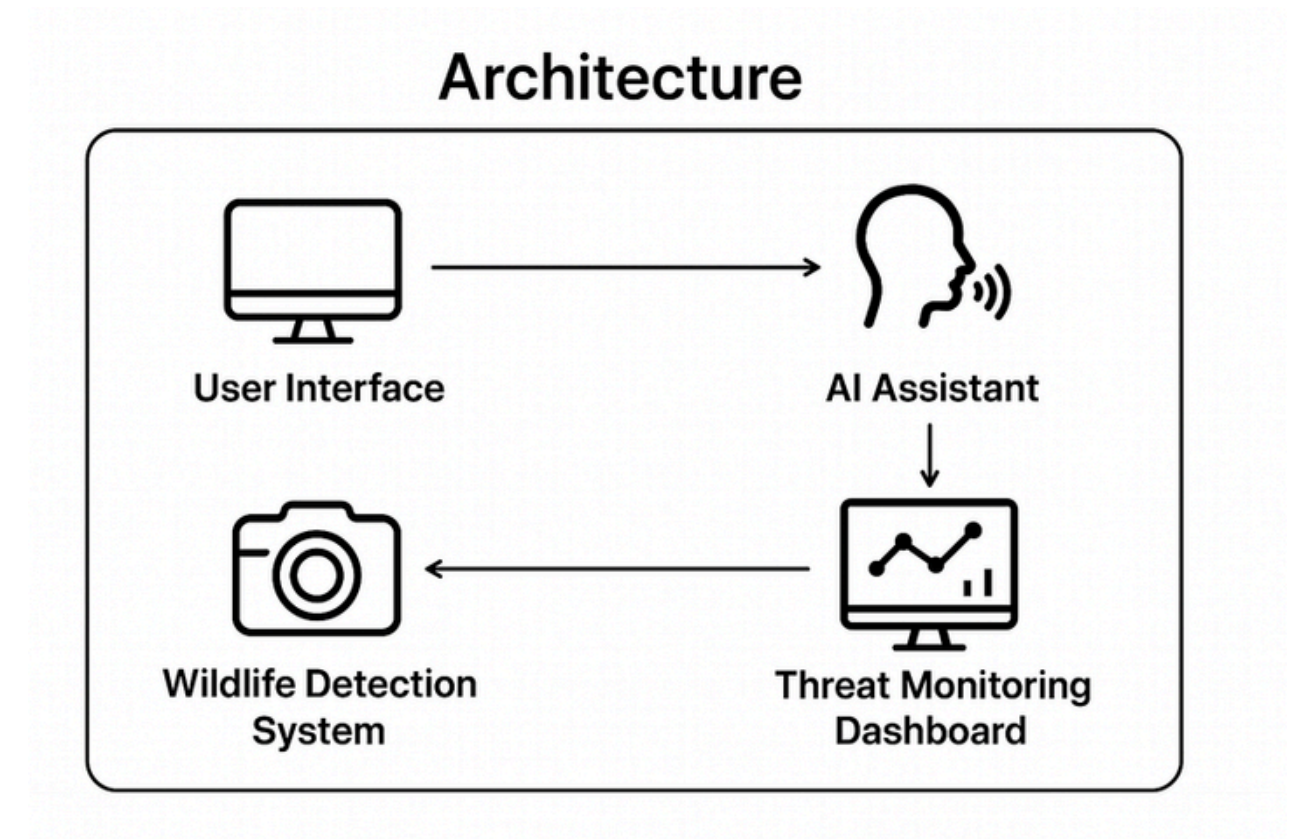
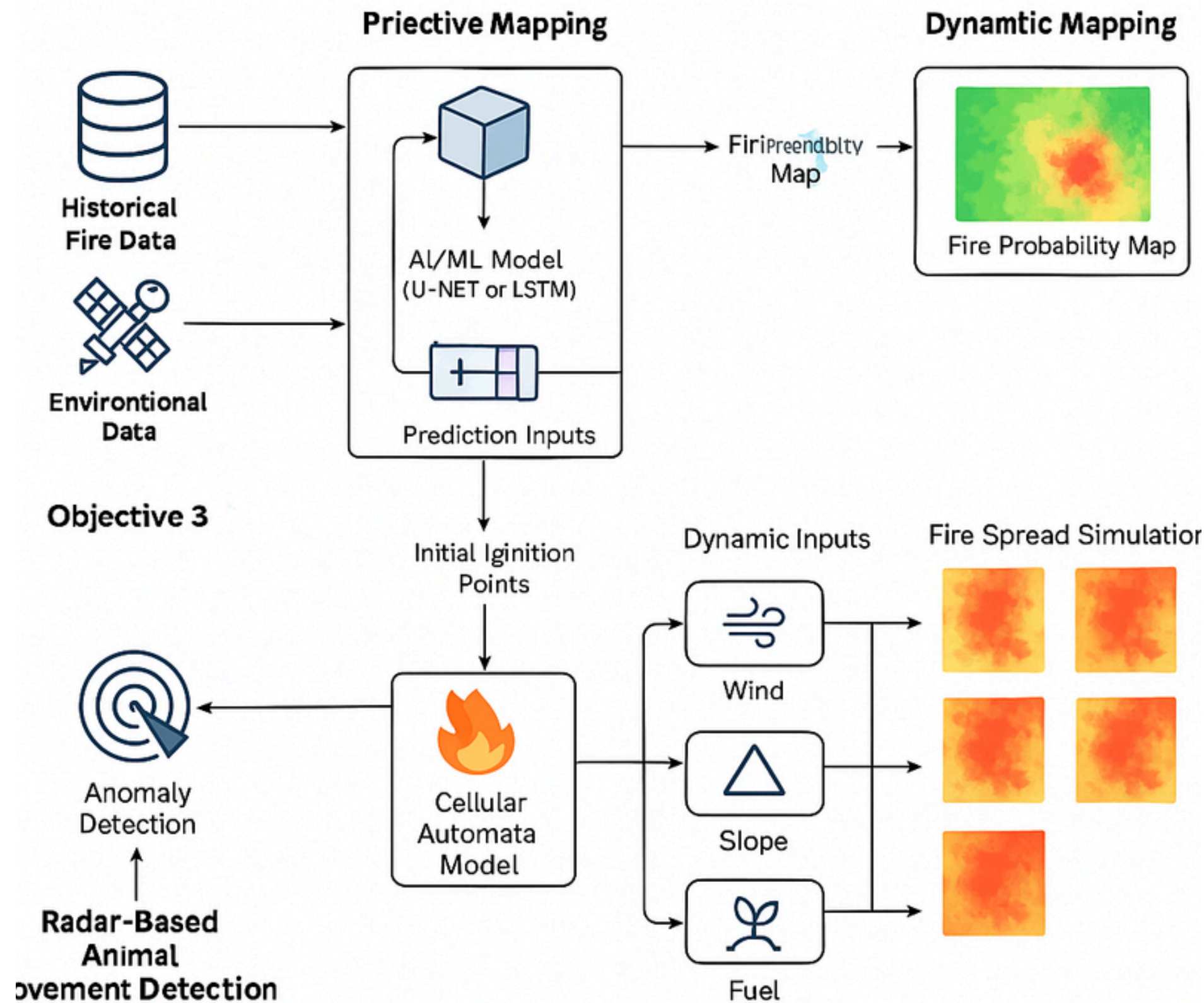
SMS Alerts

Emergency Planning



CAUTION

Architecture diagram of the proposed solution



PROJECT

SCALABILITY

Modular & Incremental Expansion

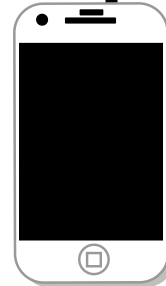
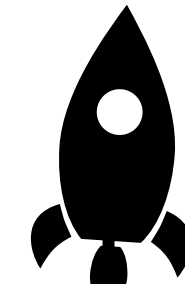
- System built as independent blocks: sensors, AI, alerts, drones
- Start with 5-10 nodes per pilot; easily add more corridors
- New sites integrate seamlessly into existing cloud dashboard

Community-Driven Cost Savings

- Local “Tech Rangers” maintain sensors & drones
- Creates employment, reduces dependence on outside technicians

Flexible Funding Model

- Supports phased rollout via government schemes, CSR, NGOs
- Step-by-step expansion without large upfront investment



Affordable Hardware & Open-Source Software

- Solar-powered motion & thermal sensors (ESP32 + LoRaWAN)
- Compact commercial drones or DIY kits
- AI & dashboards run on open-source tools (TensorFlow Lite, YOLOv8, Streamlit, OpenStreetMap)

Efficient Data Handling

- Hybrid edge-and-cloud processing: real-time alerts at edge, cloud for storage & AI updates
- Predictable operating expenses as network grows

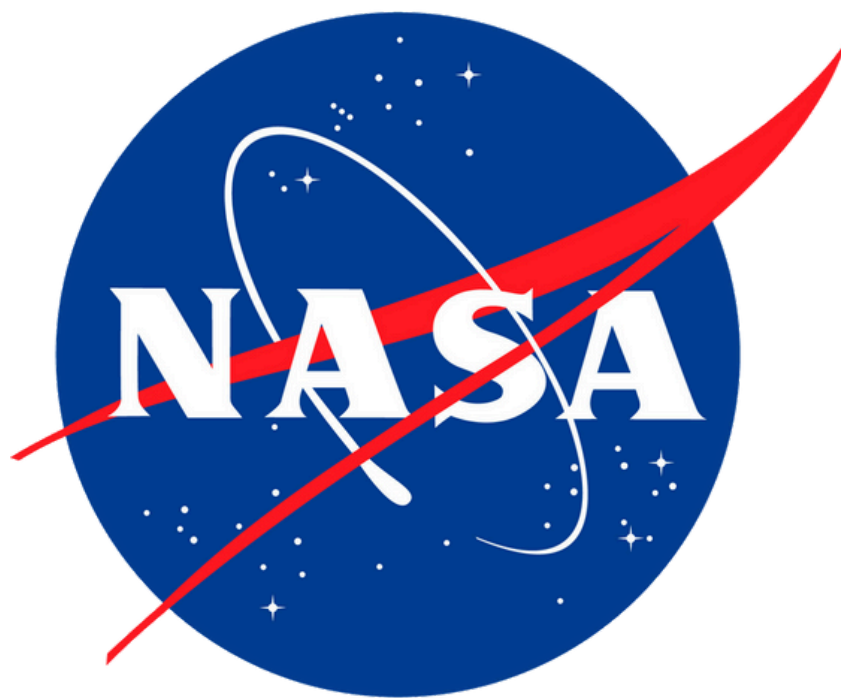
PROJECT

CONCLUSION

WORKED WITH



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(MoEFCC)



विज्ञान एवं प्रौद्योगिकी विभाग
DEPARTMENT OF
SCIENCE & TECHNOLOGY



Conclusion:

1. Statement of Novelty:

WildTrak 360 introduces a proactive approach to managing human–wildlife conflict, moving beyond the traditional reactive methods where action is taken only after damage occurs. Its novelty lies in combining low-cost solar-powered sensors, AI-based prediction, and automated drone surveillance into a single, community-driven system. Unlike scattered or manual measures, this integrated network provides advance warning, real-time tracking, and rapid response, making it both affordable and scalable for rural India.

2. Problems faced to achieve the solution and take-home message for the audience.

The system detects animal movement, predicts likely entry points, and instantly alerts villagers and forest guards through WhatsApp, SMS, and a simple mobile app, offering up to an hour's preparation time. At the same moment, drones stream live video to help rangers and trained volunteers manage the situation safely. This not only reduces losses to crops and property but also prevents harm to animals by avoiding panic-driven retaliation. Importantly, the design is built on open-source hardware and free AI tools, ensuring that communities themselves can maintain it—creating employment for local youth while strengthening conservation efforts.

On- Ground implementation

Creating a New Scientific Legacy from SPARC Foundation- TechnoTsav

Sathapathi Sai Sandeep

Science begins with wonder, scientific temperament and rational thinking. Curiosity is a basic characteristic of scientists in many fields. The ability to observe and form hypotheses to verify or refute is a foundation of the scientific method. Understanding the natural world inspired many scientists to discover and invent. The science that exists in nature brought all these inventions about. Also, due to these inventions, life has become so easy now. Consider what would happen if they did not exist. Not only this, we are sure that there will be more such inventions in the future, bringing more new things into this world. So, be curious and attentive, watch, observe, pay attention, ask questions, do the experiment, and keep inventing.

What is an incubation centre?

The incubation centre allows new-age entrepreneurs and young minds to transform innovative ideas into viable business propositions. Our primary



vision is to facilitate a platform for a budding entrepreneur to start a business venture with minimum risks.

The wait is over!

We created the Andhra Pradesh state's first scientific incubation hub at our beloved SALESIAN SISTERS schools and some local educational institutions, which are

recognised by the Andhra Pradesh state government of education.

What is the SPARC incubation centre?

SPARC (Scientific Program for Academic Research Cube) has established a Scientific Technology Incubation Centre in some districts of our state to attract and nurture young academia with innovative ideas and aptitude for carrying out research and developing the academia-industry ecosystem for new scientific technology.

Aiming to fabricate advanced engineering topics from the 5th standard onwards, SPARC's Scientific Incubation Hub collaborates with schools and college students to create synergies and synthesise scientific innovations in young minds. SPARC Foundation has launched a new scientific incubation hub innovator project. Teaching is a passion... With our scientific passion for developing engineering mindsets



Andhra Pradesh 1st scientific space tutor partners of ISRO - SPARC Innobyte

“Superintendent of Police Kakinada Shri. S. Satish kumar IPS”

Mr. Sandeep Chairman SPARC Foundation | Mr. Pradeep CEO & Director

VISION

for rural innovation



డ్రగ్స్ రహిత సమాజం

అందరి బాధ్యత

-వర్మిటీ ప్రత్యేక సదస్సులో వక్తలు

జనాదీపిక, ఎచ్చెర్ల : నిషేధిత మారక ద్రవ్యాల వినియోగం, అక్రమ రవాణాపై అందరూ కలసికట్టుగా పోరాడాలని పలువురు వక్తలు అభిప్రాయపడ్డారు. దీనివలన ఆరోగ్య రాయక సమాజం సౌధ్యమని అన్నారు. డా. డీఆర్ అంటేద్వర్ విశ్వ విద్యాలయ... జిల్లా పోలీస్ థాఖతో కలసి "సంకల్పం" కార్యక్రమాన్ని బుధవారం కేంపస్ లో నిర్వహించింది. దీనిలో భాగంగా " డ్రగ్స్ పై

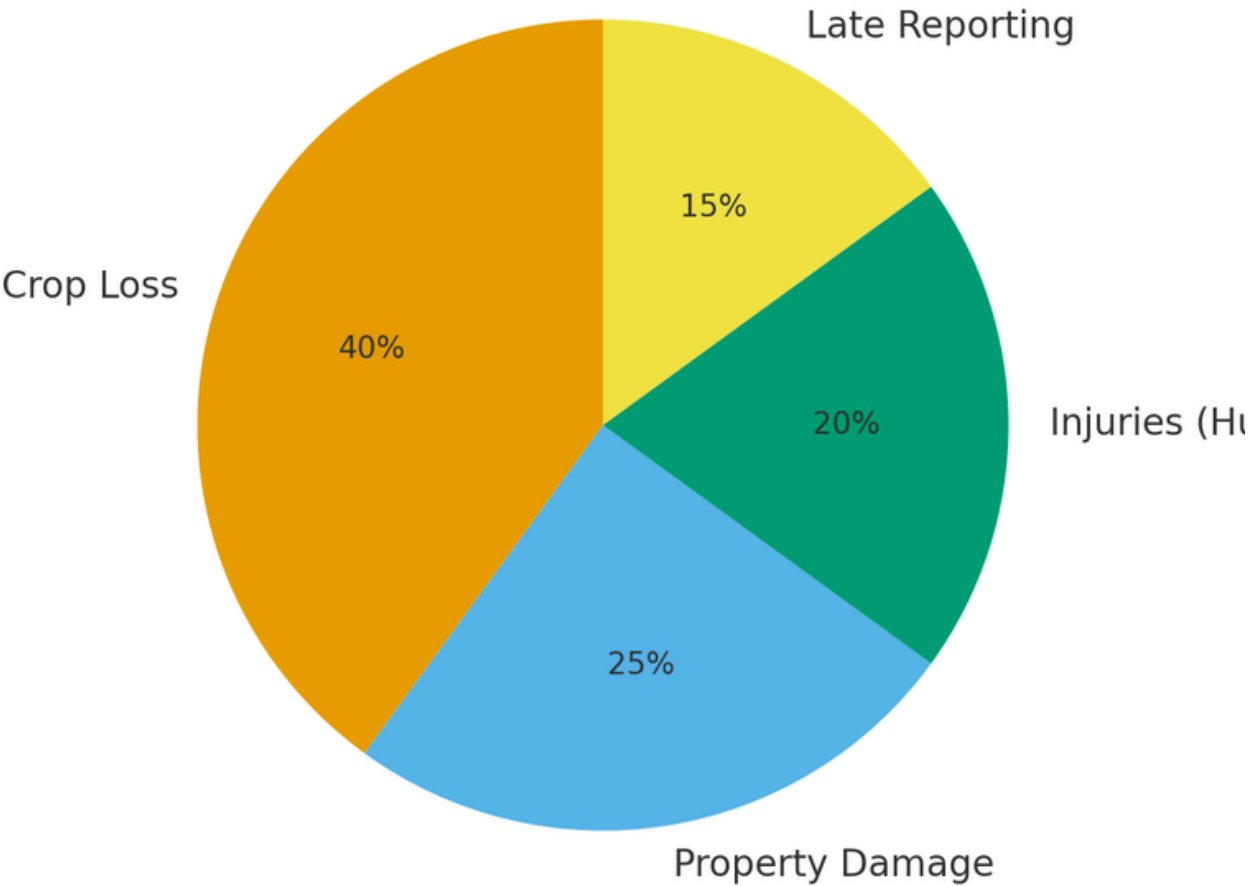
యుద్ధం : అవగాహన- మార్పు" అంశంపై ఒకరోజు సదస్సు ఆరిగింది. దీన్ని ఉద్దేశించి వర్మిటీ వైస్- ఛాన్సలర్ అచార్య కె.ఆర్. రజని మాట్లాడారు. మారక ద్రవ్యాలు వినియోగం వలన సమాజం లో మానవతా విలువలు దిగజారుతున్నాయన్నారు. డ్రగ్స్ కు దూరంగా ఉంటూ తమ లక్ష్యాలు, తల్లిదండ్రుల ఆశయాలపైనే విద్యార్థులు అధికంగా

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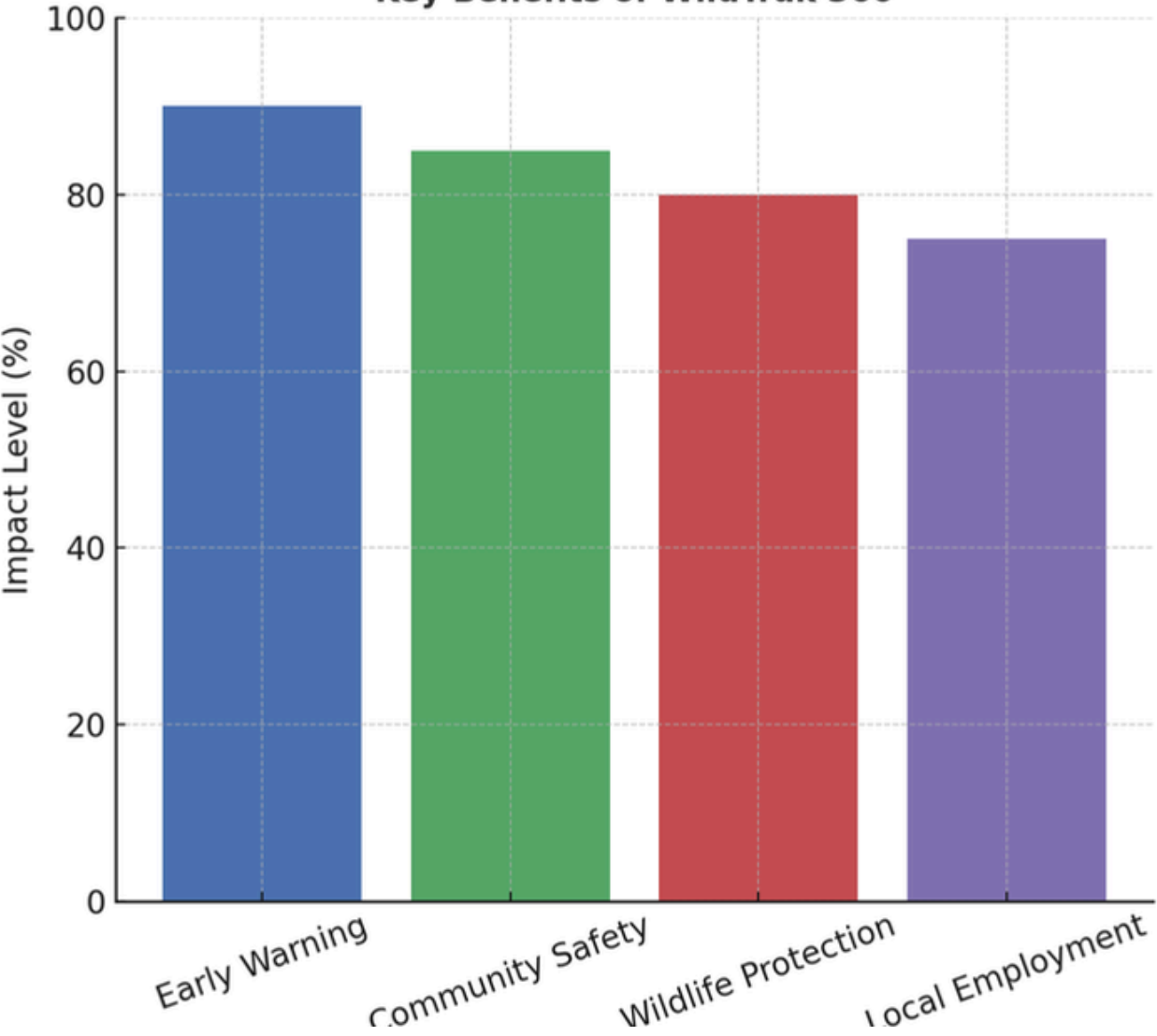
THREATS

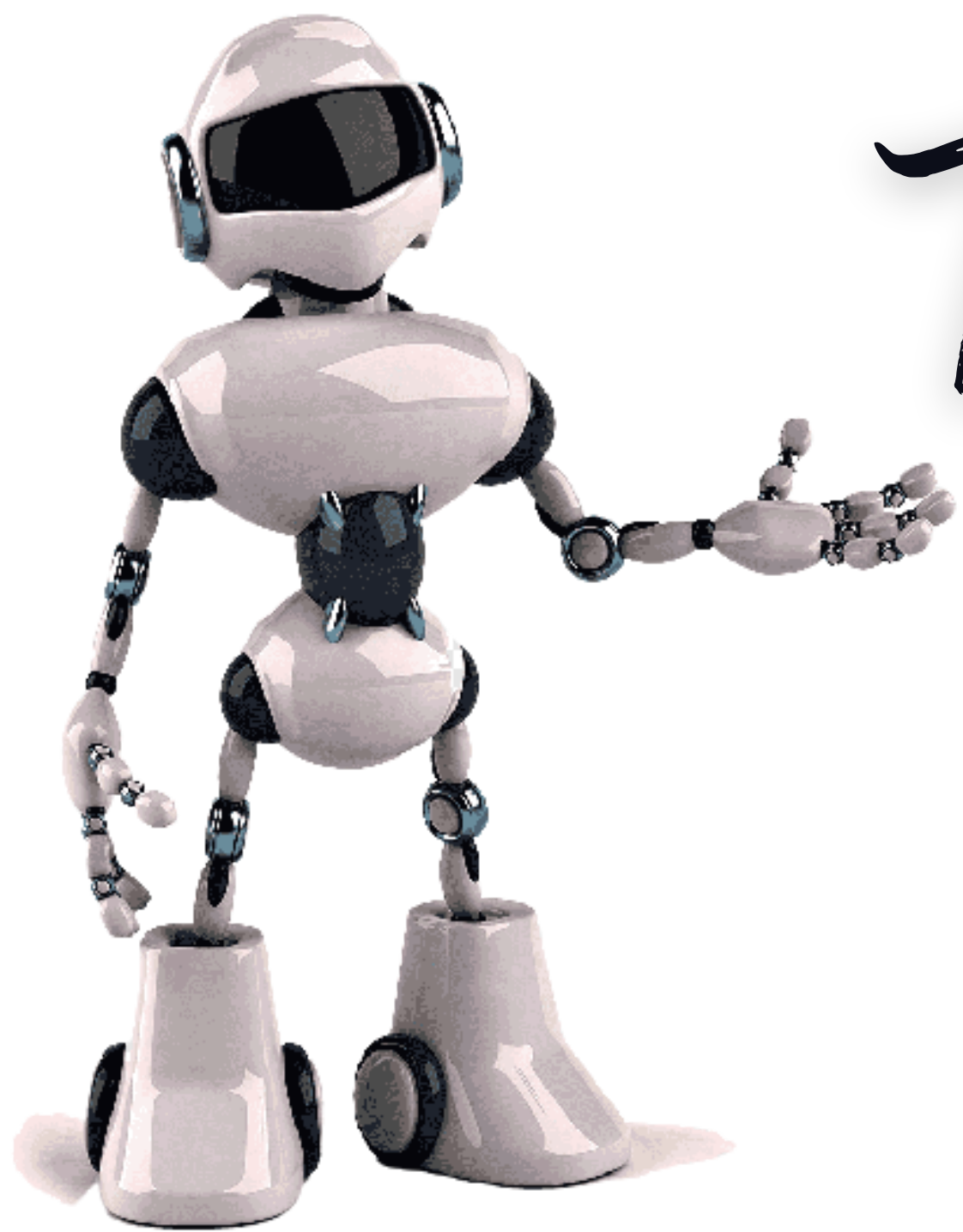
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Causes of Human-Wildlife Conflict



Key Benefits of WildTrak 360





Thank You!



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भारतीय वन्यजीव संस्थान
Wildlife Institute of India

SPARC FOUNDATION
ANDHRA PRADESH, INDIA

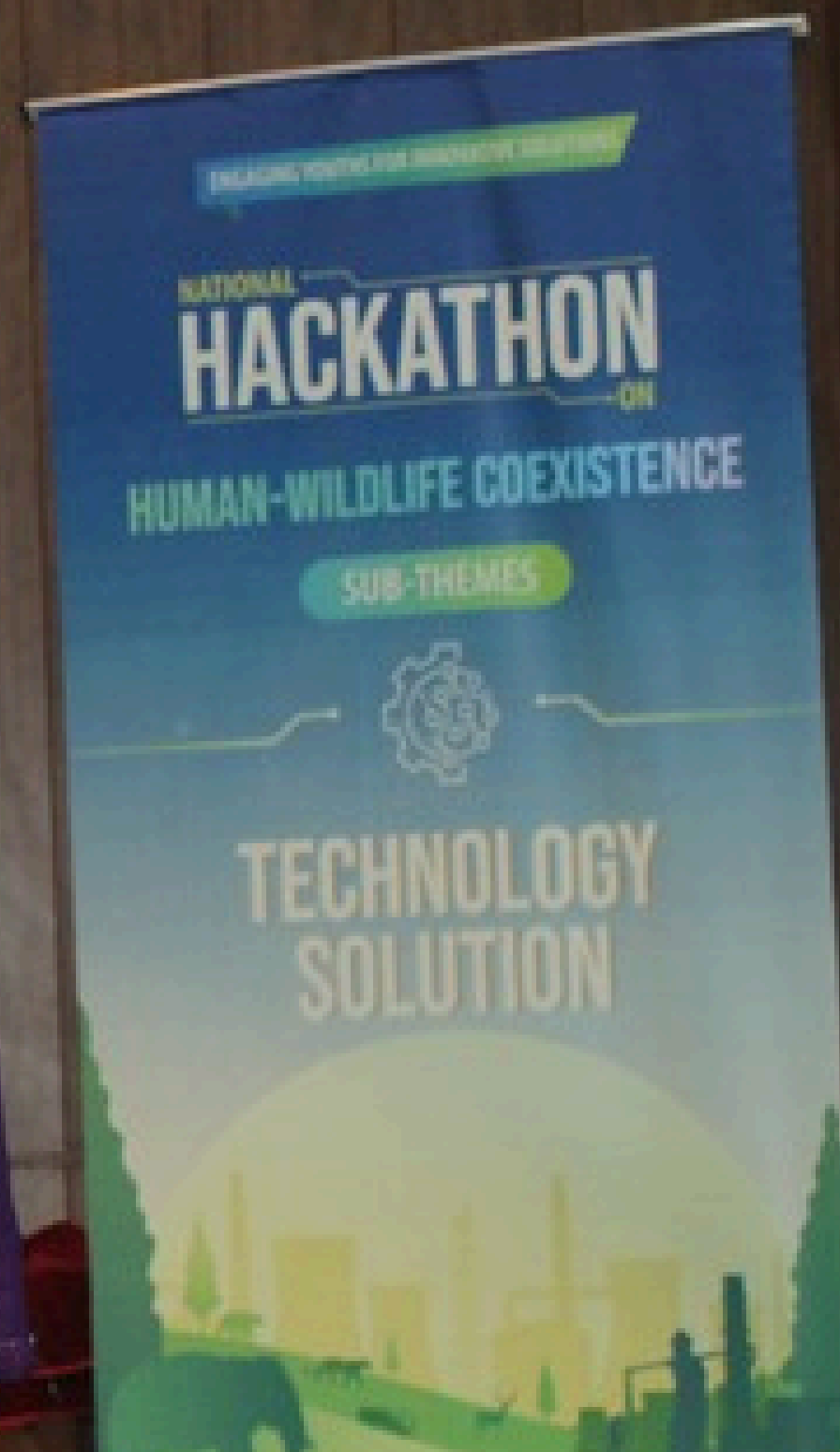
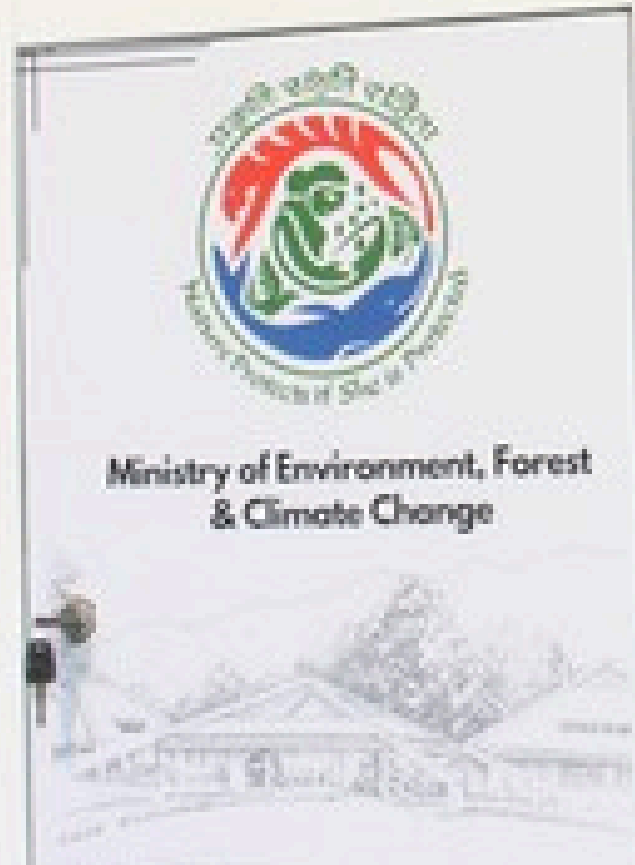


Top finalist in National Hackathon on Human wildlife - Coexistence | Dehradun

Honored by “Shri Bhupender Yadav”

Hon’ble Union Minister of Environment, Forest & Climate Change (MoEFCC)

ages, threatening both livelihoods and
move toward farms or settlements.
forest staff only after an incident has
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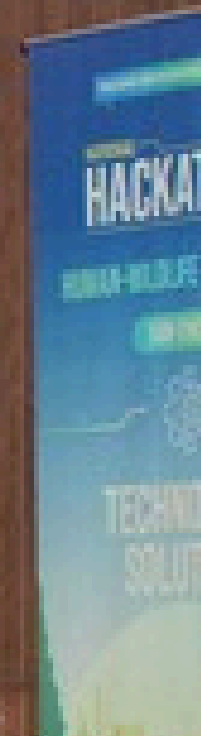
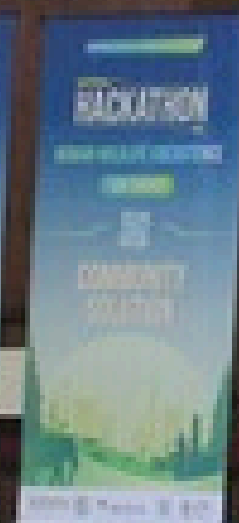
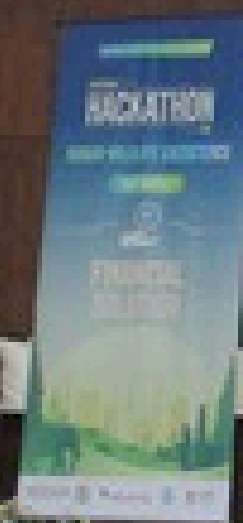
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Multilingual notifications
networks of local sirens



స్పార్క్ ఫౌండేషన్ కు జాతీయ పురస్కారం



స్పార్క్ సీఈఓ సాయిప్రదీప్ కు ప్రశంసాపత్రాన్ని
అందిస్తున్న కేంద్రమంత్రి భూపేంద్రయాదవ్

ఏలేశ్వరం: భౌగోళిక అంశాలపై వివిధ ప్రయోగాలు చేస్తున్న పట్టణానికి చెందిన స్పార్క్ ఫౌండేషన్ మంగళవారం జాతీయ పురస్కారం అందుకుంది. డెప్యూ డూన్ వైల్డ్ లైఫ్ ఇనిస్టిట్యూట్ ఆఫ్ ఇండియా ఆధ్వర్యంలో భారత ప్రభుత్వం పర్యావరణ, అరణ్య, వాతావరణ మార్పులపై జాతీయస్థాయి హ్యాక్ థాన్లో స్పార్క్ ఫౌండేషన్ ప్రశంసాపత్రాన్ని దక్కించుకుంది. దీనికి దేశవ్యాప్తంగా ఏడు జిల్లు పాల్గొనగా ఆంధ్రప్రదేశ్ నుంచి స్పార్క్ ఫౌండేషన్ విజేతగా నిలిచింది. దీంతో కేంద్రమంత్రి భూపేంద్ర యాదవ్ చేతులమీదుగా స్పార్క్ ఫౌండేషన్ సీఈఓ సాయిప్రదీప్, ప్రాజెక్టు డైరెక్టర్ సూర్యరమ్యశ్రీ ప్రశంసాపత్రం అందుకున్నారు.



కాకినాడ జిల్లా వార్తలు.. వే2న్యూస్లో. డాన్లోడ్ [way2news](https://way2news.co/sn4hpp)
way2.co/sn4hpp

యర్రవరం: స్పార్క్ సంస్థకు జాతీయ పురస్కారం

ఏలేశ్వరం మండలం యర్రవరానికి చెందిన స్పార్క్ సంస్థకు మంగళవారం జాతీయ పురస్కారం లభించింది. ఈ విషయాన్ని సంస్థ చైర్మన్ ఎస్.సాయి సందీప్ తెలిపారు. డెప్యూ డూన్లో 'మానవజాతి, వన్యప్రాణుల సహజీవనం' అంశంపై జరిగిన హ్యాక్ థాన్లో దేశం నుంచి పాల్గొన్న 120 బృందాల్లో, స్పార్క్ సంస్థ జాతీయ స్థాయిలో 7వ స్థానం దక్కించుకుంది. కేంద్ర పర్యావరణ, వాతావరణ మార్పుల శాఖ మంత్రి పురస్కారాన్ని అందించారు.

స్పార్క్ ఫౌండేషన్ సీఈవోకు ప్రశంసాపత్రం



కేంద్రమంత్రి నుంచి జాతీయ
ప్రశంసాపత్రం అందుకుంటున్న ప్రదీప్

ఏలేశ్వరం, అక్టోబరు 7 (ఆంధ్రజ్యోతి) : స్పార్క్ ఫౌండేషన్ కు జాతీయ స్థాయి గుర్తింపు లభించినట్లు మంగళవారం ఫౌండేషన్ చైర్మన్ సాయి సందీప్ తెలిపారు. భారత ప్రభుత్వం డెహరాడూన్లో అటవీశాఖ అకాడమీ నేతృత్వంలో అడవుల సంరక్షణ, పరిశోధన అంశాలపై పోటీలు నిర్వహించారు. జాతీయ స్థాయిలో ఏడు జిల్లు పాల్గొనగా ఏవీకి చెందిన స్పార్క్ ఫౌండేషన్ సీఈవో సాయిప్రదీప్ టీమ్ కు చోటుదక్కింది. కేంద్ర మంత్రి భూపేంద్ర యాదవ్ నుంచి సీఈవో ప్రదీప్ ప్రశంసాపత్రం అందుకున్నారు.